

Eight-Class Latent Class Model: Consumer Profiles for Used BEV Battery Attributes

Class 2: Range-Aware, Moderate Opt-Out		Class 3: High-Range Enthusiasts		Class 4: Degradation-Sensitive, Budget-Constrained		Class 5: Loss-Averse, Opt-Out Averse	
n = 441 • 15%		n = 466 • 16%		n = 373 • 13%		n = 596 • 20%	
Opt-Out	-11.4.		-16.3**		-34.7***	-106.0***	
Mileage (per 10k mi)	-2.3***		-2.1***		-0.6***		-3.1***
Range at yr. 3: 40-130mi (per 100mi)	26.8***		37.8		2.6**		4.9
Range at yr. 3: 130-200mi (per 100mi)	5.6		20.9***		1.9*		6.1*
Range at yr. 3: 200+mi (per 100mi)	10.5***		25.2***		1.9		5.2*
Range Loss (yr.3 - yr.8): 5-12% (per %)	-1.2***		-0.8***		-0.2**		-2.5***
Range Loss (yr.3 - yr.8): 12-24% (per %)	-1.0***		-0.7***		-0.2***		-2.6***
Range Loss (yr.3 - yr.8): 24+% (per %)	-0.7.		-0.5***		-0.2***		-2.4***
Battery Refurbishment: Pack Replace	-2.3		-0.3		-0.1		-2.5
Battery Refurbishment: Cell Replace	-2.3		-0.6		0.0		-3.8*
CLASS CHARACTERISTICS		CLASS CHARACTERISTICS		CLASS CHARACTERISTICS		CLASS CHARACTERISTICS	
Range WTP: \$26.8k pw1***, \$10.5k pw3*** Degradation: ~\$1.2k pw1***, ~\$1.0k pw2***, ~\$0.7k pw3. Moderate opt-out: 27-42% per task (mean 36%) Range anxiety 83% · Outlet 45% · Income \$94.8k EVb positive 54% · BEV ownership 7.6% EV knowledge 77% · Democratic 47%		Highest range WTP: \$37.8k / \$20.9k / \$25.2k per 100mi (all ***) Strongest degradation aversion all *** Highest income \$97.8k · Highest budget \$26.9k Most EV positive: 65.8% enviro, 35.3% functional Outlet 58% · Lowest SUV 43% · Youngest avg. 37.6 Democratic 50% · Lowest opt-out rate 2.2%		Significant degradation: **, ***, *** · Small WTP magnitudes Lowest income \$70.7k · Lowest budget \$19.9k High price sensitivity ¶ economically small WTPs ICEV-dominant · BEV ownership 3.4% Highest apartment 27% · Lowest homeowners 43% Opt-out rate 1.9% mean · Range anxiety 78%		Largest class 20% · Extreme opt-out aversion ~\$106.0k*** Strongest degradation: ~\$2.5k/~\$2.6k/~\$2.4k per % (all ***) Cell replace ~\$3.8k* · Range WTP \$4.9k-\$6.1k Highest risk-taking 45% · Income \$90.5k · Outlet 45% BEV ownership 7.8% · SUV 60% · EV positive 63% Lowest opt-out rate (mean 1.7%)	
Class 1: BEV-Skeptical, Extreme Opt-Out		Class 6: Extreme Opt-Out, Scale-Dominant		Class 7: Refurb-Averse Niche, EV-Function Skeptics		Class 8: High-Opt-Out, Budget-Constrained	
n = 230 • 8%		n = 470 • 16%		n = 169 • 6%		n = 170 • 6%	
Opt-Out	-54.2	-246.4* ¶		-69.1		-14.6**	
Mileage (per 10k mi)	-18.3		-11.1*	-5.4		-0.1	
Range at yr. 3: 40-130mi (per 100mi)	-2.9		9.7	28.6		0.5	
Range at yr. 3: 130-200mi (per 100mi)	15.3		-18.0	18.5		5.9*	
Range at yr. 3: 200+mi (per 100mi)	14.3		11.4	12.8		8.6	
Range Loss (yr.3 - yr.8): 5-12% (per %)	-7.1		0.0	0.0		-0.1	
Range Loss (yr.3 - yr.8): 12-24% (per %)	-2.1		-0.8	-2.6		-0.1	
Range Loss (yr.3 - yr.8): 24+% (per %)	0.3		-1.5.	-0.8		0.0	
Battery Refurbishment: Pack Replace	-42.5		1.6	-77.4		-1.2	
Battery Refurbishment: Cell Replace	-38.0		-5.5	-71.9		-1.5.	
CLASS CHARACTERISTICS		CLASS CHARACTERISTICS		CLASS CHARACTERISTICS		CLASS CHARACTERISTICS	
Oldest avg. age 47 · Range anxiety 88% Most anti-EV battery (27.2% enviro positive, 8.4% functional) Lowest outlet 33% · ICEV-only ~90% · SUV 67% Lowest BEV ownership 1.5% · Budget \$24.2k Extreme opt-out: 93-96% per task (mean 95.1%) 71.5% strongly disagree used BEV · Conservative 34%		Extreme opt-out ~\$246.4k* · Scale-driven effect Incoherent range WTP (pw2 negative: ~\$18.0k) Lowest range anxiety 65.6% · Most risk-taking 51% Highest BEV/PHEV: BEV 13.4%, PHEV/HEV 25.4% Highest refuel frequency 5.5/mo · Budget \$28.8k Most diverse: white 60%, African American 22%		Smallest class 6% · Extreme refurb: ~\$77.4k/~\$71.9k (no sig) Most EV-function negative (87.5% do not disagree) Moderate range WTP \$18.5k-\$28.6k (not significant) Loss pw2 ~\$2.6k · Opt-out rate 15-21% (mean 18%) SUV 64% · Budget \$30.4k · BEV ownership 11% All WTPs statistically insignificant		Highest opt-out rate 49-60% per task (mean 55%) Opt-out aversion ~\$14.6k** but low attribute response Lowest income \$68.2k · Lowest budget \$18.9k Range anxiety 85% · Outlet 37% · ICEV-heavy 81% BEV ownership 4.7% · Renters 46% 46.6% strongly disagree used BEV purchase	

WTP in \$1,000 USD. Piecewise linear range and range loss. Bar lengths capped at ~\$120k for display; ¶ indicates truncation with actual value shown. Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1. N = 2,916. Cars & SUVs combined.